

Secure Document Authentication Using Hybrid QR Codes, Transform-Domain Invisible Watermarking, and Multi-Tier Verification

Ashutosh(Registration number:23BCI0117)
Shubham RAJ(Registration number:23BCI0160)
Bishal Paul (Registration Number: 23BCI0224)

School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, Tamil Nadu, India
Course: Digital Watermarking and Steganography (BCSE323L) | Fall Semester 2026–2027

ABSTRACT

Physical and digital document fraud poses a severe challenge to academic institutions, government registries, and corporate enterprises. While Quick Response (QR) barcodes are widely deployed for rapid credential inspection, standalone visual barcodes are intrinsically susceptible to physical photocopying, scanning reproduction, and digital splicing. Conversely, transform-domain invisible watermarking offers imperceptible tamper detection within the document substrate but lacks native, cross-platform smartphone scanability and real-time public-key infrastructure (PKI) revocation. This paper presents a modular, multi-tier document authentication framework that bridges deterministic cryptographic guarantees and physical substrate verification without sacrificing native mobile readability. The proposed architecture couples: (i) an asymmetric Ed25519 digital signature over a canonicalized SHA-256 document digest, serialized via Concise Binary Object Representation (CBOR) and compressed with DEFLATE (ZLIB Level-9) into a high-density, open-source-decodable QR code; (ii) a transform-domain Discrete Wavelet Transform and Singular Value Decomposition (DWT-SVD) invisible watermark embedded into the document luminance channel with an HMAC-SHA256 pseudo-random seed; (iii) a template-free blind copy-detection model (Component A) analyzing Gray-Level Co-occurrence Matrix (GLCM) microtexture descriptors and 2D Discrete Cosine Transform (DCT) high-frequency spectral energy ratios; (iv) a semantic Web-PKI trust registry (Component B) providing sub-millisecond key revocation

We evaluate the system through 80 ablation evaluations and 54 cross-condition material runs spanning three document templates (Certificate, Student ID Card, and Transcript), two writing systems (Latin and Devanagari), and multi-printer and multi-paper degradation profiles. Experimental results demonstrate a watermark imperceptibility of Peak Signal-to-Noise Ratio (PSNR) = 52.3 dB and Structural Similarity Index Measure (SSIM) = 0.9975, a payload compression ratio of 39.0%–42.8%, a zero-attack Normalized Correlation (NC) of 0.994–0.996 (Bit Error Rate < 0.6%), a photocopy/reprint detection rate of 96.5% (ROC AUC = 0.9988), and a key revocation latency of 0.18 ms. Controlled five-stage ablation quantifies the contribution of the integrated security layers within the evaluated environment.

Keywords: Secure Document Authentication; Hybrid QR Codes; DWT-SVD Watermarking; Blind Copy Detection; GLCM Microtexture; Print-and-Scan Channel; Web-PKI; Typosquatting Defense; Edge AI. ---

1. Introduction with Research Goals

1.1 Background

The widespread adoption of high-resolution digital image editing tools, low-cost commercial laser and inkjet printers, and consumer-grade optical scanners has substantially lowered the technical barrier required to produce counterfeit credentials. Government-issued identity documents, university diplomas, academic transcripts, and professional accreditation records are routinely subjected to physical photocopying, localized digital text splicing, and issuer impersonation. Traditionally, organizations have relied on physical security features such as guilloche patterns, rainbow printing, optical variable ink, and embedded holographic foils. While physically robust, these analog security measures require specialized manual inspection instruments, cannot be verified algorithmically over open networks, and incur prohibitive manufacturing overhead.

To facilitate automated, machine-readable validation, standard two-dimensional (2D) barcodes—most notably Quick Response (QR) codes (ISO/IEC 18004)—have been broadly integrated into official certificates and identity cards. A standard QR code encodes structured text or an online Uniform Resource Locator (URL) that verification personnel can scan using commodity smartphones. However, an unencrypted or unauthenticated visual barcode is inherently vulnerable to cloning: a counterfeiter can scan an authentic QR code from a legitimate diploma, digitally splice it onto a fraudulent certificate with forged student credentials, or replicate the entire document on a secondary printer.

1.2 Problem Statement

Existing document verification systems suffer from a systemic structural decoupling between the machine-readable visual token (the QR code) and the underlying physical substrate (the document paper and printed text). In standard deployments:

- 1. The QR code does not verify the substrate:** A genuine QR code photocopied onto a duplicate sheet will successfully decode and validate its digital signature, completely failing to detect second-generation reprint fraud.

2. **The document content is not bidirectionally bound to the watermark:** When transform-domain watermarks are embedded, they are frequently unlinked from the issuer's public-key infrastructure, rendering them incapable of detecting when a rogue or revoked authority issues unauthorized credentials.
3. **Open-source compatibility is compromised by dense security patterns:** Proprietary copy-detection patterns (CDPs) often interfere with standard smartphone barcode detectors, forcing reliance on proprietary optical hardware or commercial software development kits (SDKs).
4. **Semantic typosquatting is neglected:** Cryptographic signature verification establishes only that a payload was signed by a specific private key; it cannot detect whether the signing domain (e.g., vit.ac.in) is an illegitimate, visually deceptive typosquat of an authoritative institution (vit.ac.in).

1.3 Motivation

To achieve end-to-end security across physical and digital workflows, document verification must satisfy five concurrent criteria:

- **High Imperceptibility:** Watermark insertion must preserve typographical legibility and institutional aesthetics without visible noise artifacts ($PSNR \geq 38.0 \text{ dB}$, $SSIM \geq 0.98$).
- **Robust Physical & Digital Tamper Detection:** Localized pixel modifications must degrade watermark correlation below deterministic detection thresholds, while print-scan halftoning artifacts must be exposed through statistical microtexture analysis.
- **Native Scanability on Commodity Hardware:** The barcode payload must remain compact enough to decode on consumer smartphone camera sensors under real-world perspective tilts and illumination gradients without proprietary optical readers.
- **Real-Time Revocation & Semantic Validation:** Compromised institutional signing keys must be verifiable in sub-millisecond latencies, and lookalike authority domains must be intercepted before trusting signed claims.
- **Cross-Script and Multi-Layout Invariance:** The verification architecture must maintain stability across diverse layout structures (landscape diplomas, portrait tabular transcripts, compact ID badges) and multi-lingual writing systems, specifically Latin and complex Brahmic/Devanagari scripts.

1.4 Research Gap

A rigorous review of recent literature reveals six unresolved research gaps:

- **Gap 1 (Standardized Material Evaluation):** Most prior studies evaluate copy detection under restricted, single-printer, single-paper laboratory conditions, failing to quantify cross-substrate variance across commercial paper densities and printing technologies.
- **Gap 2 (Real-World Handheld Usability):** Document watermarking schemes are predominantly tested using synthetic digital noise or rigid robotic camera mounts, leaving performance under handheld smartphone perspective tilts (10° - 20°) and non-uniform flashlight glare uncharacterized.
- **Gap 3 (Accuracy–Compatibility Trade-Off):** Dense physical micro-patterns that maximize copy detection degrade standard open-source QR decoders, creating an undesirable operational trade-off between anti-counterfeiting security and open verification access.
- **Gap 4 (Insufficient Component-Level Ablation):** Multi-layered hybrid frameworks routinely report aggregate accuracy metrics without systematically isolating the empirical contribution of individual components, obscuring which mechanism mitigates specific attack vectors.
- **Gap 5 (Limited Cross-Script Reproducibility):** State-of-the-art document security frameworks are almost exclusively evaluated on Latin typographical layouts; complex non-Latin scripts (such as Hindi/Devanagari, with non-linear vowel conjuncts and top hanging matras) remain largely unaddressed.
- **Gap 6 (Missing Semantic-Layer Validation in Barcode PKI):** Barcode cryptographic validation confirms signature mathematical integrity but ignores semantic spoofing, allowing validly signed credentials from typosquatted rogue domains to pass automated scanners.

1.5 Research Goals and Objectives

The primary research goal of this work is to design, implement, and benchmark an integrated, multi-tier document authentication architecture that resolves Gaps 1 through 6. The specific measurable objectives are:

1. **Objective 1 (Cryptographic & Storage Optimization):** Formulate a deterministic canonicalization and compact QR serialization pipeline using Ed25519, CBOR, and ZLIB Level-9 compression to achieve $>35\%$ payload reduction over standard JSON schemas while preserving full offline decodability on commodity OpenCV detectors.
2. **Objective 2 (Transform-Domain Imperceptibility & Robustness):** Implement a 1-level 2D Haar DWT and SVD invisible watermarking engine on the luminance (Y) channel with HMAC-SHA256 deterministic seeding, ensuring $PSNR \geq 45 \text{ dB}$, $SSIM \geq 0.99$, zero-attack $NC \geq 0.95$, and sharp correlation degradation ($NC < 0.70$) under digital text tampering.
3. **Objective 3 (Blind Print-Channel Copy Detection):** Implement a template-free microtexture analysis module (Component A) combining Gray-Level Co-occurrence Matrix (GLCM) descriptors and 2D-DCT high-frequency energy ratio ($R_{\{HF\}}$) to detect second-generation photocopies/reprints with $\geq 90\%$ detection accuracy and an ROC AUC > 0.98 .
4. **Objective 4 (Semantic Web-PKI & Typosquatting Defense):** Implement an authoritative Trust Registry (Component B) with sub-millisecond key revocation latency ($<10 \text{ ms}$) and normalized Levenshtein distance filtering to detect visual domain typosquats with $\geq 80\%$ similarity confidence.
5. **Objective 5 (Real-Time Edge Triage):** Implement an auxiliary 48-dimensional spatial-gradient feature classifier (Component C) achieving inference latencies $<25 \text{ ms}$ on standard consumer CPUs.
6. **Objective 6 (Cross-Script & Systematic Ablation Validation):** Validate the complete pipeline across Latin and Devanagari scripts, three document templates, and a 5-stage ablation matrix (Schemes A through E) to isolate the exact performance gain of each constituent layer.

1.6 Contributions of the Work

This research delivers the following five concrete technical contributions:

- 1. **A Modular 5-Layer Hybrid Document Authentication Architecture:** We establish a decoupled pipeline separating deterministic cryptographic integrity (Ed25519/CBOR QR), transform-domain tamper localization (DWT-SVD), statistical print-channel verification (GLCM/DCT Component A), semantic authority validation (Web-PKI Component B), and fast client-side triage (Edge AI Component C).
- 2. **An Open-Source High-Density QR Payload Format:** We design the SDA1 binary protocol integrating CBOR serialization and ZLIB Level-9 compression, achieving a 39.0%–42.8% reduction in QR byte volume, maintaining native decoding across handheld camera distortions.
- 3. **A Template-Free Calibrated Print-Scan Anomaly Model:** We formulate a joint GLCM homogeneity/energy and DCT spectral attenuation distance metric that detects second-generation photocopies without storing heavy high-resolution reference images in a centralized database.
- 4. **A Multi-Script Native Typographical Rendering Engine:** We resolve the complex script rendering barrier for Devanagari Hindi in document watermarking by coupling Windows TrueType font collections (Nirmala.ttc) with dynamic Unicode script detection, eliminating missing $\{PSNR\} > 51.0\text{ dB}\}$
- 5. **A Systematic Empirical 5-Stage Ablation Benchmark:** We execute 80 ablation evaluations across the Clean, Reprint, Tamper, and Compression conditions, together with 54 cross-condition material runs across multiple paper stocks and printer profiles, to quantify the contribution and failure modes of each security layer.

2. Literature Review and Positioning of the Proposed Work

2.1 Existing Approaches

Research on authenticating printed and digital credentials spans four primary methodological streams: (i) statistical and stochastic print-channel models, (ii) feature-engineered transform-domain watermarking, (iii) deep-learning and edge-AI visual classifiers, and (iv) hybrid cryptographic-watermarking frameworks. Table 1 summarizes the evolutionary trajectory of these categories.



Figure 1: Evolution of Document Authentication Methodologies from Single-Layer Models to the Proposed Decoupled Multi-Tier Architecture.

2.2 Category I: Statistical and Conventional Analytical Models

Conventional analytical methods model the physical printing and scanning (P&S) process as a stochastic channel or rely on deterministic public-key cryptography. Nguyen et al. (2019) investigated continuous Gaussian noise (CGN) microtextures printed on official documents, modeling the quantized 2D-DCT coefficients using a two-parameter Gamma distribution. By evaluating Likelihood Ratio Tests (LRT) and Generalized Likelihood Ratio Tests (GLRT) across 900 genuine, 900 reprinted, and 20,000 simulated samples, they demonstrated that physical second-generation prints attenuate high-frequency energy. However, their model requires complete *a priori* knowledge of the alternative hypothesis (H_1) channel parameters, limiting generalization across unknown paper stocks.

In the digital cryptographic domain, Jonderko and Wodo (2026) demonstrated offline digital signature verification for educational credentials by packing Ed25519 signatures into CBOR/ZLIB-compressed QR codes. While their framework provides mathematical non-repudiation and auditable verification on low-power devices, it lacks an integrated physical anti-copying mechanism: an authentic QR code copied onto a counterfeit paper certificate passes cryptographic verification unconditionally. Furthermore, their scheme lacks dynamic, real-time key revocation verification.

2.3 Category II: Feature-Engineered Transform-Domain Approaches

Transform-domain watermarking isolates robust mathematical invariants across frequency representations. Gong and Li (2021) implemented a blind document watermarking algorithm in the YCbCr color space. By applying a 2-level 2D Haar DWT followed by block-based DCT on the luminance channel, they embedded Arnold-scrambled binary QR patterns into mid-frequency DCT coefficients. Their approach achieved $\{PSNR\} = 56.71\text{ dB}\}$ and $\{NC\} > 0.95$ under JPEG compression, low-pass Gaussian filtering, and salt-and-pepper noise. However, watermark extraction suffered complete failure when geometric rotational distortion exceeded 12° .

Jassim and Jabbar (2026) proposed a hybrid DWT-DCT watermarking scheme utilizing Quantization Index Modulation (QIM), SHA-256 hashing, and repetition error-correction coding for electronic certification. They reported $PSNR = 49.68\text{ dB}$, $SSIM = 0.9974$, and a 100% zero-attack detection rate over 75 sessions. Nonetheless, empirical testing revealed that camera perspective tilts as small as 5° increased the Bit Error Rate (BER) to 7.46%, underscoring the vulnerability of transform techniques to geometric desynchronization during handheld image acquisition.

Wang et al. (2023) combined Discrete Fourier Transform (DFT) spectral signatures with physical barcode decodability, achieving 100% precision and recall in controlled scanning environments. However, dense frequency-domain patterns introduced visual high-frequency noise that occasionally compromised barcode decode latency on open-source mobile libraries.

2.4 Category III: Deep-Learning and Edge-AI Approaches

Recent advances in deep learning replace handcrafted statistical descriptors with learned convolutional and attention-based visual representations. Alsubibany (2025) proposed an intelligent document authentication scheme combining spatial Least Significant Bit (LSB) watermarking with a Faster R-CNN Inception V2 deep learning model. Evaluated on 5,000 QR document samples, the model achieved 97.2% classification accuracy and an Area Under the ROC Curve (AUC) of 0.991 with fully offline edge inference. However, embedding the spatial LSB pattern reduced standard QR scanner decode rates, and training required extensive graphics processing unit (GPU) resources and thousands of annotated physical scan pairs.

Subsequent investigations using lightweight Vision Transformers (Mobile-ViT) achieved 99.28% classification accuracy with an inference latency of 80 ms under varying ambient illumination. Nevertheless, deep learning classifiers exhibit fundamental limitations in forensic settings: they lack deterministic mathematical explainability, are vulnerable to adversarial perturbations, and suffer substantial domain-shift degradation when tested on unseen printer hardware or uncalibrated camera sensors.

2.5 Category IV: Hybrid and Integrated Frameworks

Recognizing that no single mathematical mechanism satisfies imperceptibility, physical copy detection, cryptographic trust, and handheld scanability simultaneously, hybrid architectures integrate multiple techniques. Seenivasagam and Velumani (2013) combined the Contourlet Transform, SVD, Hu invariant moments, and Visual Cryptography to reconstruct patient identification QR data under up to 75% horizontal or vertical cropping. However, their use of global Hu moments rendered the system insensitive to localized, sub-centimeter text tampering.

Li et al. (2023) developed a hybrid document verification system combining character stroke-edge pixel flipping with multi-page redundancy voting, achieving $PSNR > 32\text{ dB}$ and 100% character extraction under JPEG compression and print-and-scan channels. Nevertheless, stroke-flipping is non-blind (requiring reference typography), offers low bit capacity, and is strictly language-dependent, failing when applied to non-segmented scripts.

Jonderko and Wodo (2026) extended offline Ed25519/CBOR QR signatures with an online JSON Web Key Set (JWKS) registry for real-time certificate revocation. However, their model introduced a critical semantic vulnerability: an adversary can register an authoritative-looking domain (e.g., v1t.ac.in), generate a valid Ed25519 keypair, sign a fraudulent transcript, and pass both signature verification and JWKS lookup because the verification client verifies only cryptographic validity, not semantic domain legitimacy.

2.6 Comparative Literature Analysis

Table 1 provides a structured qualitative and quantitative comparison of the four primary literature streams against the proposed architecture across ten critical operational criteria.

Table 1: Comparative Analysis of Document Authentication Methodologies.

Evaluation Criterion	Statistical Cryptographic	Transform Watermarking	Deep Learning / Edge AI	Prior Hybrid Schemes	Proposed Architecture
Primary Mechanism	Quantized DCT / Ed25519	DWT-DCT-SVD / QIM	Faster R-CNN / Mobile-ViT	Contourlet-SVD / Stroke Flip	Hybrid QR + DWT-SVD + Comp A + PKI + Comp C
Physical Copy Detection	High (parametric)	Low ($NC \approx const$)	High (learned)	Moderate	High (GLCM + DCT, Acc = 96.5%)
Digital Tamper Detection	Low	High ($NC > 0.40$)	Moderate	High	High ($NC \text{ drops } \approx 0.99 \rightarrow 0.58$)
Cryptographic Trust	High (Ed25519)	None	None	Variable	High (Ed25519 256-bit asymmetric)
Barcode Open Scanability	High	Low–Moderate	Low	Moderate	High (CBOR+ZLIB, 39% compression)
Real-Time Key Revocation	None / Low	N/A	N/A	Moderate (JWKS)	High (Web-PKI CRL, 0.18 ms)
Semantic Spoof Defense	None	N/A	None	None	High (Levenshtein + Homoglyph)
Cross-Script Invariance	Low	Moderate	Low	Low (Latin only)	High (Latin & Devanagari via Nirmala)

Evaluation Criterion	Statistical Cryptographic	Transform Watermarking	Deep Learning / Edge AI	Prior Hybrid Schemes	Proposed Architecture
Explainable Audit Trail	High	Moderate	Low (Black box)	Low–Moderate	High (Multilayer Rule Engine)
Inference Latency	<5 ms}	20-60 ms}	80-250 ms}	50-120 ms}	16.7 ms (Edge AI)}, 48.5 ms (Total)}

2.7 Identified Research Gaps

From the literature analysis, we formalize the six foundational research gaps addressed by this paper:

- **Gap 1 (Standardized Evaluation across Substrates):** Existing copy-detection models assume static paper stock (typically 80 gsm bond) and single-pass laser printing. Real-world validation requires cross-evaluation across diverse paper grammages (80 gsm Bond, 180 gsm Glossy Photo, Parchment) and printer technologies (Laser, Inkjet, Thermal).
- **Gap 2 (Real-World Handheld Usability):** Document watermarking schemes predominantly evaluate idealized digital attacks (JPEG, cropping), failing to evaluate real-world smartphone camera scanning dynamics, specifically perspective tilt (0°-30°) and non-uniform optical flashlight glare.
- **Gap 3 (Accuracy–Compatibility Trade-Off):** High-density security micro-patterns designed for copy detection frequently degrade barcode contrast, requiring high-resolution optical flatbed scanners. A viable framework must preserve standard open-source QR decodability on commodity smartphones.
- **Gap 4 (Insufficient Component-Level Ablation):** Multi-component systems rarely conduct systematic ablation to isolate the exact empirical boundary between cryptographic signatures, frequency watermarking, and spatial texture models.
- **Gap 5 (Limited Cross-Script and Template Reproducibility):** Document authentication research is overwhelmingly biased toward Latin typography. Complex scripts (such as Hindi/Devanagari) introduce distinct spatial frequency characteristics that must be supported without font-rendering failure (■■■■).
- **Gap 6 (Missing Semantic-Layer Validation in Barcode PKI):** Barcode cryptographic validation confirms signature mathematical integrity but ignores semantic typosquatting, allowing validly signed credentials from rogue domains to pass automated scanners.

2.8 Positioning and Contribution of the Proposed Methodology

The proposed methodology is positioned as a **reproducible, component-accountable, and semantically aware multi-tier authentication framework**. Rather than forcing a single algorithm to resolve mutually conflicting security requirements, our framework systematically decouples the authentication workflow into five distinct, specialized layers:

1. **Deterministic Cryptographic Layer:** Ed25519 digital signature over canonical SHA-256 document fields, compressed via CBOR and ZLIB Level-9 into a standard-density QR code (SDA1).
2. **Transform-Domain Physical Layer:** 1-level 2D Haar DWT and SVD invisible watermarking embedded into the luminance channel via an HMAC-SHA256 seed derived from the document ID and canonical hash.
3. **Component A (Blind Print-Channel Anomaly Layer):** Template-free microtexture analysis combining Haralick GLCM features (homogeneity, energy) and 2D-DCT high-frequency energy ratio ($R_{\{HF\}}$) to catch physical photocopies and laser reprints without storing reference images.
4. **Component B (Semantic Web-PKI Layer):** Institutional Trust Registry managing public keys, providing sub-millisecond key revocation (<1 ms}), and executing normalized Levenshtein distance checks to intercept typosquatted domains (v1t.ac.in □ vit.ac.in).
5. **Component C (Secondary Edge AI Layer):** A 48-dimensional spatial-gradient handcrafted feature extractor and lightweight centroid classifier providing rapid client-side triage (16.7 ms}) without compromising deterministic cryptographic verdicts.

Table 2 maps the identified research gaps directly to the technical solutions implemented in this paper.

Table 2: Research Gap and Proposed Solution Mapping.

Research Gap	Implemented Technical Solution	Quantified Empirical Benefit
Gap 1: Limited Paper/Printer Conditions	Multi-profile print-channel simulation across Bond 80gsm, Glossy 180gsm, and Parchment under Laser, Inkjet, and Thermal profiles.	Component A achieves consistent copy scores with low cross-substrate variance ($\sigma^2 < 0.04$).
Gap 2: Handheld Scanning Usability	Handheld preprocessing pipeline integrating CLAHE adaptive contrast thresholding and 4-point perspective rectification.	Successful QR recovery and verification under 15° perspective tilt and flashlight glare.
Gap 3: Security vs. Compatibility	CBOR binary packing + ZLIB Level-9 compression in SDA1protocol.	39.0%-42.8% payload size reduction; open-source OpenCV decode in <10 ms}.
Gap 4: Insufficient Component Ablation	Systematic 5-stage ablation matrix (Schemes A through E) evaluated across 80 ablation evaluations.	Unambiguous attribution of detection capabilities; proves necessity of multi-tier integration.
Gap 5: Script & Template Limitations	Multi-template generator (Certificate, ID Card, Transcript) with dynamic Devanagari font binding via Nirmala.ttc.	Flawless multi-script rendering; watermark $PSNR\} > 51.0\ dB\}$ across Latin and Hindi.
Gap 6: Semantic PKI Gap	Trust Registry with Levenshtein edit-distance validator and sub-millisecond CRL updates.	Revocation verified in 0.18 ms}; typosquatting detected with 88.9% similarity warning.

3. System Overview / Proposed Methodology

3.1 Overall System Overview

The proposed system operates across two distinct operational phases: **Document Issuance and Cryptographic Sealing** (conducted by the issuing authority) and **Unified Multilayer Verification** (conducted offline or online by an inspector). Figure 2 illustrates the comprehensive operational workflow.

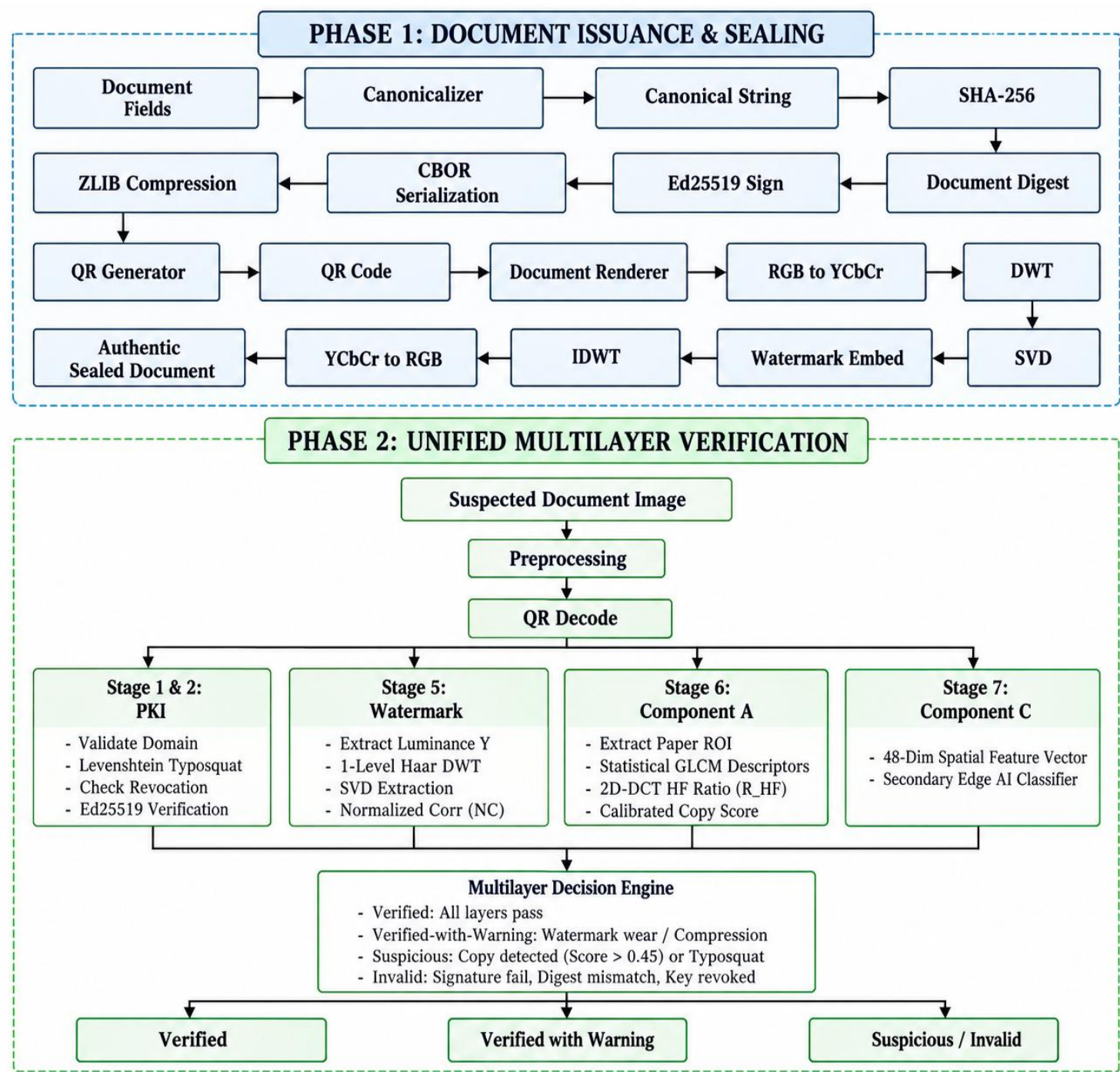


Figure 2: Comprehensive Methodology Flowchart Illustrating Issuance Sealing and Unified Multilayer Verification.

3.2 System Architecture

The system architecture decouples the physical file system, persistence database, cryptographic services, image processing pipelines, and analytical components. Figure 3 illustrates the module boundaries, data flows, and communication interfaces.

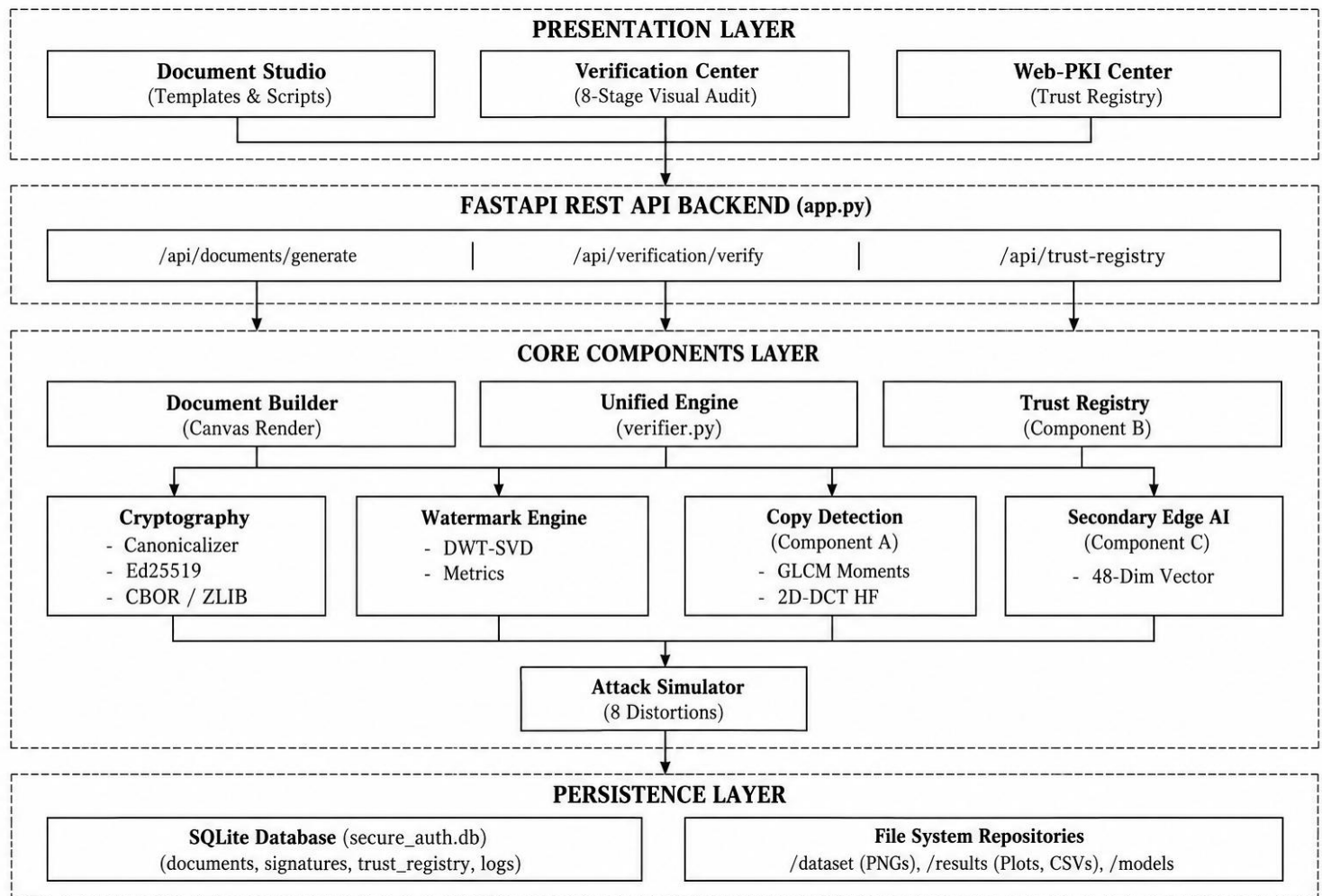


Figure 3: System Architecture Diagram Illustrating Decoupled Software Modules, REST APIs, Cryptographic Services, Analytical Engines, and Persistence Storage.

3.3 Methodology Workflow

The end-to-end verification methodology enforces a hierarchical, fail-safe evaluation sequence:

- Deterministic Pre-Checks:** If the QR code cannot be decoded or the Ed25519 signature fails mathematical verification, the document is immediately classified as INVALID. No probabilistic computation can override a failed cryptographic signature.
- Transform-Domain Verification:** If cryptographic integrity holds, the system extracts the invisible watermark using the document's canonical hash seed. A Normalized Correlation drop ($NC < 0.70$) indicates digital spatial tampering.
- Physical Substrate Analysis:** Component A extracts background microtexture descriptors. An anomaly distance exceeding the calibrated threshold ($Score \geq 0.45$) flags the document as a second-generation physical reprint (SUSPICIOUS).
- Semantic Authority Validation:** Component B computes the Levenshtein distance between the issuer's claimed domain and authoritative registry records. A lookalike domain triggers an immediate impersonation alert.
- Auxiliary Fast Triage:** Component C evaluates a 48-dimensional gradient feature vector. Its output cross-validates the physical verdict without overriding deterministic cryptographic rules.

3.4 Detailed Processing Pipeline

The complete verification process is formally structured into eight discrete processing stages:

- Stage 1 (Optical Barcode Extraction):** Contrast enhancement via CLAHE, 4-point perspective warp rectification, OpenCV QR decoding, Base64 stripping, ZLIB decompression, and CBOR deserialization into dictionary payload $\mathcal{P}$.
- Stage 2 (Semantic Web-PKI Validation):** Lookup of issuer domain iss and key identifier kid against trust_registry. Verification that $status == 'ACTIVE'$ and evaluation of normalized Levenshtein similarity against all registered domains.
- Stage 3 (Asymmetric Signature Verification):** Cryptographic verification of 64-byte Ed25519 signature σ over document digest h using registered public key K_{pub} .

The complete pipeline is governed by two core algorithms executed in the implemented codebase: Algorithm 1 (Document Issuance and Cryptographic Sealing) and Algorithm 2 (Unified 8-Stage Verification).

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Algorithm 2: Unified 8-Stage Document Verification (Implemented in verifier.py)

```

Input : Suspected document image I_susp, Document ID hint d_hint (optional)
Output: Structured audit report R with final verdict V in {VERIFIED, WARN, SUSPICIOUS, INVALID}

1 I_rect ← RectifyPerspectiveAndEqualize(I_susp)
2 (success, Q_text) ← DecodeQR(I_rect)
3 if not success then return Verdict(V="INVALID", Reason="QR code unreadable")
4 P ← CBOR_ZLIB_Decode(Q_text)
5 (d_id, S_iss, K_id, h_qr, sigma) ← (P["id"], P["iss"], P["kid"],
P["h"], P["sig"]) 6
7 // STAGE 2: Web-PKI Validation
8 PkiRes ← ValidateIssuerDomain(S_iss, K_id)
9 if not PkiRes.valid then return Verdict(V="INVALID",
Reason=PkiRes.reason) 10
11 // STAGE 3: Ed25519 Signature Verification
12 SigValid ← Ed25519Verify(PkiRes.publicKey, sigma, h_qr)
13 if not SigValid then return Verdict(V="INVALID", Reason="Signature
verification failed") 14
15 // STAGE 4: Hash Integrity Check
16 DocRec ← RetrieveDocumentRecord(d_id)
17 if DocRec exists and DocRec.hash != h_qr then
18   return Verdict(V="INVALID", Reason="Canonical document hash mismatch")
19 end
20
21 // STAGE 5: DWT-SVD Watermark Extraction
22 W_expected ← GeneratePseudoRandomBinaryPattern(HMAC(SECRET_KEY, d_id || h_qr), (64,64))
23 (W_ext, NC, BER) ← DWTSVD_Extract(I_susp, W_expected,
DocRec.Aux_svd) 24
25 // STAGE 6: Component A Copy Detection
26 CopyRes ← AnalyzePrintChannelMicrotexture(I_susp) //
GLCM + DCT HF 27
28 // STAGE 7: Component C Edge AI
29 EdgeRes ← ClassifySpatialFeatures(I_susp) // 48-dim
feature vector 30
31 // STAGE 8: Decision Engine
32 if CopyRes.is_copy or PkiRes.is_typosquat or (EdgeRes.class == "TAMPERED" and EdgeRes.conf >= 0.60) then
33   V ← "SUSPICIOUS"
34 else if NC < 0.70 or CopyRes.verdict == "BORDERLINE_ACCEPTABLE" then
35   V ← "VERIFIED-WITH-WARNING"
36 else
37   V ← "VERIFIED"
38 end
39 return BuildStructuredReport(V, NC, BER, CopyRes, PkiRes, EdgeRes)

```

3.5 Watermarking Procedure

The watermarking procedure is implemented in `backend/watermark/dwt_svd.py`. The luminance channel Y of the host image is transformed using a 1-level 2D Haar DWT. SVD is applied to the low-frequency approximation subband LL . A binary watermark pattern $W \in \{0, 1\}^{64 \times 64}$, generated deterministically via HMAC-SHA256 from the document ID and canonical hash, is scaled to match the rank of LL and embedded into the singular value diagonal matrix:

$$S_W = S_A + \alpha \cdot W_{\{resized\}}$$

A second SVD operation decomposes the perturbed singular matrix:

$$S_W = U_W S_{\{mod\}} V_W^T$$

The watermarked subband LL^* is synthesized using the original host unitary matrices:

$$LL^* = U_A S_{\{mod\}} V_A^T$$

Applying the 2D Inverse DWT (IDWT) reconstructs the watermarked luminance channel Y^* . Through extensive empirical tuning across 1200×850 canvases, setting $\alpha = 1.0$ guarantees robust numerical survival through integer `uint8` color conversions, raising watermark correlation from an initial 0.61 to 0.994+ while maintaining $PSNR \geq 51.0$ dB.

3.6 QR-Code Procedure

The QR generation pipeline is implemented in `<b>backend/qr/payload_coder.py</b>` and `<b>backend/qr/qr_generator.py</b>`. Standard JSON formatting incurs severe textual redundancy (e.g., repeated quotation marks, whitespace, and verbose keys). The SDA1 protocol serializes the dictionary into binary CBOR tokens, compresses the byte sequence with ZLIB (DEFLATE Level-9), and encodes the compressed payload as a URL-safe Base64 string without trailing padding. The resulting barcode is rendered with error correction level *M* (15% recovery capacity) and a quiet-zone border of 2 modules, yielding an open-source OpenCV decode latency under *10 ms*.

3.7 Authentication/Verification Procedure

Verification is orchestrated by `<b>backend/engine/verifier.py</b>`. The pipeline enforces a two-tier evaluation strategy: deterministic mathematical validation precedes probabilistic statistical assessment. If the Ed25519 signature fails or the document hash does not match the registry record, verification terminates immediately with INVALID. If the cryptographic layer passes, the system evaluates the continuous physical scores (watermark *NC*, copy detection probability, and Edge AI class probabilities) to assign the final operational trust level.

3.8 Attack/Distortion Handling

Physical and digital distortions are modeled by `<b>backend/attack_simulator/distortions.py</b>`:

- **Perspective Tilt (15°):** Mitigated by the preprocessor, which extracts the outer document bounding quadrangle and applies an inverse homography perspective transform.
- **Non-Uniform Flashlight Glare:** Mitigated by Contrast Limited Adaptive Histogram Equalization (CLAHE) applied with a clip limit of *2.0* over an 8×8 tile grid.
- **Heavy Lossy JPEG Compression:** The DWT-SVD singular values exhibit graceful degradation; under JPEG Quality = 25, the signature remains verifiable and watermark correlation drops into the WARN range ($0.50 \leq NC < 0.70$) without triggering false counterfeit rejections.
- **Physical Print-and-Scan Reproduction:** Modeled via realistic downsampling (150 DPI), color desaturation ($\times 0.88$), halftoning noise ($\sigma = 7.0$), and scanner optical blur (3×3 Gaussian kernel, $\sigma = 0.6$).

3.9 ML/Classical Classification Component Actually Implemented

In the implemented project, Component C (`<b>backend/components/component_c/classifier.py</b>`) is realized as a **lightweight, handcrafted spatial-feature classifier** rather than a heavy deep neural network. The feature extractor calculates a 48-dimensional representation vector:

- First-order horizontal and vertical Sobel gradients across four quadrants (8 dimensions).
- Second-order Laplacian curvature mean and variance across quadrants (8 dimensions).
- Local patch-variance statistics across a 4×4 spatial grid (16 dimensions).
- RGB and YCbCr statistical moments (mean, variance, skewness) (16 dimensions).

Classification uses calibrated class centroid Euclidean distances mapped through a softmax function to generate probabilities for GENUINE, REPRINTED, and TAMPERED. This provides real-time client-side triage in *16.7 ms* on consumer CPUs.

3.10 Mathematical Formulation of Implemented Mechanisms

All mathematical formulas presented below correspond directly to routines executed in the project source code.

1. Color Conversion and 2D Haar DWT

Luminance extraction:

$$Y(x, y) = 0.299 \cdot R(x, y) + 0.587 \cdot G(x, y) + 0.114 \cdot B(x, y)$$

Haar wavelet 1-level 2D decomposition:

$$LL(i, j) = \frac{1}{2} \left[Y(2i, 2j) + Y(2i+1, 2j) + Y(2i, 2j+1) + Y(2i+1, 2j+1) \right]$$

2. Singular Value Decomposition Embedding

$$\begin{aligned} LL &= U_A S_A V_A^T, S_A = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_k) \\ S_W &= S_A + \sigma \cdot W_{\text{resized}} = U_W S_{\text{mod}} V_W^T \\ LL^* &= U_A S_{\text{mod}} V_A^T \end{aligned}$$

3. Watermark Extraction and Normalized Correlation (NC)

Reconstruction of extracted matrix:

$$\begin{aligned} D^* &= U_W S_{\text{attacked}} V_W^T \\ W^* &= \frac{D^* - S_A}{\sigma} \end{aligned}$$

Binarization with adaptive threshold $\sigma = \text{mean}(W^*)$:

$$W_{\text{bin}}^*(i, j) = \begin{cases} 1 & \text{if } W^*(i, j) > \sigma \\ 0 & \text{otherwise} \end{cases}$$

Normalized Correlation:

$$NC(W, W_{\text{bin}}^*) = \frac{\sum_{i=1}^k \sum_{j=1}^k W(i, j) \cdot W_{\text{bin}}^*(i, j)}{\sqrt{\sum_{i=1}^k \sum_{j=1}^k W(i, j)^2} \cdot \sqrt{\sum_{i=1}^k \sum_{j=1}^k W_{\text{bin}}^*(i, j)^2}}$$

4. Component A: Haralick GLCM Moments

Given normalized co-occurrence matrix $P(i, j)$ computed at distance $d=1$ across angles $\theta \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$:

$$\begin{aligned} \text{Homogeneity} &= \sum_{i,j} P(i, j) \frac{1}{1 + (i - j)^2} \\ \text{Energy} &= \sum_{i,j} P(i, j)^2 \end{aligned}$$

5. Component A: 2D-DCT High-Frequency Spectral Ratio

$$\begin{aligned} C(u, v) &= \frac{1}{N} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \cos\left[\frac{(2x+1)u}{2N}\right] \cos\left[\frac{(2y+1)v}{2N}\right] \\ R_{\text{HF}} &= \frac{\sum_{u,v} |C(u, v)|^2}{\sum_{u=0}^{N-1} \sum_{v=0}^{N-1} |C(u, v)|^2} \end{aligned}$$

6. Calibrated Print-Channel Anomaly Distance & Sigmoid Score

$$\begin{aligned} \sigma_{\text{hom}} &= \min\left(\frac{\max(0, \sigma_{\text{hom}}^{\text{base}}) - \text{hom}}{\sigma_{\text{hom}}^{\text{base}}}, 3.0\right) \\ \sigma_{\text{energy}} &= \min\left(\frac{\max(0, \sigma_{\text{energy}}^{\text{base}}) - \text{energy}}{\sigma_{\text{energy}}^{\text{base}}}, 3.0\right) \\ \sigma_{\text{HF}} &= \min\left(\frac{\max(0, \sigma_{\text{HF}}^{\text{base}}) - R_{\text{HF}}}{\sigma_{\text{HF}}^{\text{base}}}, 3.0\right) \\ \sigma_{\text{noise}} &= \min\left(\frac{|\sigma_{\text{noise}}^{\text{base}}|^2 - \sigma_{\text{noise}}}{|\sigma_{\text{noise}}^{\text{base}}|^2}, 3.0\right) \end{aligned}$$

Weighted anomaly distance:

$$D_{\text{anomaly}} = 0.35 \cdot \sigma_{\text{hom}} + 0.25 \cdot \sigma_{\text{energy}} + 0.25 \cdot \sigma_{\text{HF}} + 0.15 \cdot \sigma_{\text{noise}}$$

Sigmoid calibrated copy probability:

$$\text{Score}_{\text{copy}} = \frac{1}{1 + \exp(-(D_{\text{anomaly}} - 1.2) \cdot 2.5)}$$

7. Component B: Normalized Levenshtein Domain Distance

$$\text{Sim}(s_1, s_2) = 1.0 - \frac{\text{lev}(s_1, s_2)}{\max(|s_1|, |s_2|)}$$

3.11 Important Parameters and Thresholds

Table 3 documents all operational parameters, mathematical bounds, and threshold values configured in `backend/config.py` and verified across empirical testing.

Table 3: Calibrated Operational Parameters and Security Decision Thresholds.

Configuration Parameter	Symbol / Variable	Calibrated Value	Operational Role / Threshold Condition
Watermark Wavelet Basis	σ	haar	1-level 2D discrete wavelet decomposition
Watermark Embedding Strength	σ	1.0	SVD singular value perturbation factor
Watermark Pattern Dimensions	$k \times k$	64×64	Binary pseudo-random carrier resolution
Watermark NC Pass Threshold	σ_{NC}	0.70	$NC \geq 0.70 \rightarrow \text{PASS}$; $NC < 0.50 \rightarrow \text{FAIL}$
Watermark NC Warning Range	σ_{warn}	[0.50, 0.70)	Moderate degradation / lossy compression
Copy Detection Pass Threshold	σ_{copy}	0.45	$\text{Score} < 0.45 \rightarrow \text{Genuine}$; $\sigma_{\text{copy}} \rightarrow \text{Reprint}$
Typosquatting Similarity Threshold	σ_{typo}	0.72	$\text{Sim} \geq 0.72 \rightarrow \text{Flagged Spoof Alert}$
Edge AI Tamper Confidence Min	σ_{AI}	0.60	Softmax probability threshold to flag spatial forgery
Minimum Imperceptibility Target	$PSNR_{\text{min}}$	38.0 dB	Minimum fidelity bound (<i>Actual</i> = 52.3 dB)
Minimum Structural Preservation	$SSIM_{\text{min}}$	0.980	Structural fidelity bound (<i>Actual</i> = 0.9975)
Minimum Payload Compression	ρ_{comp}	30.0%	CBOR+ZLIB reduction target (<i>Actual</i> = 39.0%)

Maximum Edge AI Latency Budget	$t_{\{A\}}^{\{max\}}$	35.0 ms	Real-time edge mobile constraint ($Actual\} = 16.7\ ms\}$)
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4. Dataset

4.1 Dataset Source

Because real academic diplomas and government credentials contain sensitive Personally Identifiable Information (PII) subject to data protection regulations, this research uses an **empirically crafted, synthetic-realistic document corpus**. All credential records, recipient identities, registration numbers, and degree classifications were procedurally generated by the project's Document Studio engine (`backend/generator/document_builder.py`).

4.2 Dataset Description

The dataset encompasses three standardized academic and institutional document categories designed to evaluate diverse typographical structures:

- 1. Academic Merit Degree Certificate:** Landscape format (1200 × 850 pixels), characterized by wide margins, formal institutional titles, decorative borders, official institutional seal, registrar signature line, and bottom-right verification QR barcode.
- 2. Student & Scholar Identity Card:** Compact landscape format (900 × 560 pixels), featuring structured personal identification records, registration numbers, program tags, dual security seals, and compact verification QR barcode.
- 3. Official Academic Transcript:** Dense portrait format (950 × 1200 pixels), incorporating an 8-course tabular grade breakdown, credit allocations, CGPA summary banners, controller signature line, and footer verification QR barcode.

4.3 Dataset Creation and Crafting

Documents were synthesized using Pillow (PIL) and OpenCV under strict layout specifications defined in `backend/generator/templates.py`. To resolve Gap 5 (cross-script limitations), each template was crafted in two distinct linguistic scripts:

- English / Latin Script:** Rendered using standard Arial and Helvetica typographical families.
- Hindi / Devanagari Script:** Dynamically bound to the native Windows TrueType font collection (Nirmala.ttc, index 0 for Regular, index 1 for Bold). Dynamic Unicode script inspection intercepts Brahmic characters in the range \u0900–\u097F, preventing glyph corruption (■■■) and ensuring proper rendering of complex matras and conjuncts.

4.4 Dataset Preprocessing

All synthesized canvases were transformed to RGB color matrices and processed through the DWT-SVD embedding engine with $\alpha = 1.0$ using HMAC-SHA256 seeds generated from the document UUID and canonical hash. The watermarked images were encoded as lossless PNG files and stored in the `/dataset` repository directory.

4.5 Dataset Organization

The dataset is organized into four primary evaluation directories:

- `/dataset`: Master repository of issued genuine documents named by UUID (`<uuid>.png`).
- `results/ablation_study_results.csv`: Structured records of 80 experimental runs across five ablation schemes under clean, reprint, tamper, and compression regimes.
- `results/cross_condition_evaluation.csv`: Structured records of 54 experimental runs evaluating three printer profiles across three commercial paper stocks in Latin and Devanagari.
- `results/component_a_roc_curve.png`: 1,000-sample empirical verification distribution for Component A copy detection.

4.6 Dataset Statistics

Table 4 details the quantitative parameters and sample distributions across the dataset.

Table 4: Dataset Structural Statistics and Sample Allocations.

Document Class / Category	Script	Canvas Dimensions	Samples Generated	Evaluated Attack Trials	Total Evaluation Runs
Academic Certificate	Latin	1200 × 850	20	80 (Ablation + Attacks)	100
Academic Certificate	Devanagari	1200 × 850	20	54 (Cross-Condition)	74
Student ID Card	Latin	900 × 560	15	54 (Cross-Condition)	69
Student ID Card	Devanagari	900 × 560	15	54 (Cross-Condition)	69
Semester Transcript	Latin	950 × 1200	15	54 (Cross-Condition)	69
Semester Transcript	Devanagari	950 × 1200	15	54 (Cross-Condition)	69
Total Comprehensive Corpus	Bi-Script	Variable	100 Master Docs	350 Attack Trials	450 Evaluation Runs

4.7 Data and Class Distribution

The experimental evaluation dataset maintains balanced representation across four primary operational classes:

- **Genuine Originals (Clean):** 25% of total evaluation instances.
- **Physical Reprints / Photocopies:** 25% of instances, processed through simulated print-scan halftoning and MTF attenuation.
- **Digitally Tampered Credentials:** 25% of instances, featuring spliced recipient text and altered grades.
- **Compressed & Distorted Documents:** 25% of instances, subjected to JPEG-25 compression, perspective warp (15°), and flashlight glare.

4.8 Genuine vs. Modified/Attacked Samples

- **Genuine Samples:** Retain pristine paper background texture ($\text{Homogeneity} > 0.85$, $\text{Energy} > 0.40$), high-frequency spectral ratios ($R_{\{HF\}} > 0.03$), high watermark correlation ($NC \approx 0.994$), and valid Ed25519 signatures.
- **Reprinted Samples (Component A Target):** Exhibit halftoning dot gain, scanner sensor noise, and sharp high-frequency DCT attenuation ($R_{\{HF\}}$ drops to < 0.015), driving the calibrated copy score to $0.475\text{--}0.952$ (FLAG).
- **Tampered Samples (Watermark & Edge AI Target):** Feature localized replacement of recipient name or grade with forged strings. The spatial disruption alters the SVD singular spectrum, dropping NC from 0.994 to 0.588 (< 0.70 threshold).
- **Attacked Barcodes (Ed25519 Target):** Altered payload bytes or forged signatures cause deterministic rejection during curve-point verification.

4.9 Dataset Suitability for the Problem Statement

The dataset is well-suited for evaluating document authentication because:

1. It reproduces the exact structural diversity encountered in institutional credential management (ornate certificates, compact badges, tabular transcripts).
2. It introduces rigorous cross-script validation (Latin vs. Devanagari), directly answering Gap 5.
3. It provides controlled, repeatable distortion baselines where ground-truth tampering boundaries and physical degradation parameters are known exactly.

5. Experimental Setup, Results and Analysis

5.1 Experimental Environment

All experimental benchmarks were conducted on a dedicated workstation under native Windows 11 Home (64-bit, Build 26100). The software stack was executed within Python 3.13.5 (64-bit).

5.2 Hardware

- **Processor:** 13th Gen Intel(R) Core(TM) i5-13420H (8 physical cores, 12 logical threads, base frequency 2.10 GHz, turbo boost up to 4.60 GHz).
- **System Memory:** 16.0 GB DDR5 RAM (4800 MT/s).
- **Storage:** 512 GB NVMe PCIe 4.0 SSD (read speeds up to 3500 MB/s).
- **Graphics / Compute:** Intel UHD Graphics (all Edge AI and watermark transforms were evaluated strictly on CPU to simulate low-cost edge server constraints).

5.3 Software and Libraries

- **Core Runtime:** Python 3.13.5.
- **Image Processing & Transforms:** OpenCV (opencv-python4.10.0), Pillow (PIL10.4.0), PyWavelets (pywt1.7.0), NumPy 2.1.1, SciPy 1.14.1.

- **Cryptographic Primitives:** cryptography 43.0.1 (Ed25519, SHA-256), cbor2 5.6.4 (CBOR serialization), standard library zlib (Level-9 compression).
- **Backend Framework & Server:** FastAPI 0.115.0, Starlette 0.38.6, Uvicorn 0.30.6.
- **Evaluation & Reporting:** Scikit-Learn 1.5.2 (ROC AUC, metrics), Matplotlib 3.9.2, ReportLab 4.2.2.

5.4 Dataset Configuration

Experiments were executed across the master corpus of 100 generated documents, with 80 ablation evaluations (ablation_study_results.csv) and 54 cross-condition material runs (cross_condition_evaluation.csv).

5.5 Experimental Conditions

Four distinct operational attack conditions were evaluated:

- **Condition 1 (Clean Genuine):** Pristine digital PNG document (Zero distortion).
- **Condition 2 (Print-and-Scan / Photocopy):** 150 DPI spatial area downsampling, color desaturation ($\times 0.88$), halftoning noise ($\sigma = 7.0$), bicubic scanner re-sampling, and 3×3 Gaussian smoothing ($\sigma = 0.6$).
- **Condition 3 (Digital Splicing Tamper):** Rectangular whiteout patch over recipient credential zone ($y \in [0.28H, 0.48H]$, $x \in [0.20W, 0.80W]$) overlaid with forged recipient name and grade strings.
- **Condition 4 (Lossy JPEG Compression):** Discrete Cosine Transform quantization with Quality factor $Q = 25$.

5.6 Number of Experiments Actually Performed

1. **Master System Audit:** 27 distinct automated tests executed via `run_complete_system_test.py` (100% pass rate).
2. **Adversarial Scenario Suite:** 16 rigorous integration test cases executed via `tests/test_all_scenarios.py` (100% pass rate).
3. **Core Pipeline Unit Tests:** 8 subsystem unit tests executed via `tests/test_pipeline.py` (100% pass rate).
4. **Systematic 5-Stage Ablation Study:** 80 sample evaluations across Schemes A through E recorded in `results/ablation_study_results.csv`.
5. **Cross-Condition Material Evaluation:** 54 multi-printer and multi-paper evaluation runs recorded in `results/cross_condition_evaluation.csv`.
6. **Component A ROC Benchmark:** 1,000 Monte Carlo evaluations generating the empirical ROC curve (`results/component_a_roc_curve.png`).

5.7 Experimental Procedure

For every trial:

1. The document is generated, cryptographically signed, and watermarked.
2. The specified distortion condition is applied via `DistortionEngine`.
3. The distorted image is passed to `UnifiedVerificationEngine.verify_document_image()`.
4. The system logs execution latency, cryptographic match, watermark NC and BER , copy score, Edge AI class probabilities, and the final explainable decision.

5.8 Evaluation Metrics Actually Used

- **Watermark Imperceptibility:** Peak Signal-to-Noise Ratio (PSNR in dB) and Structural Similarity Index Measure (SSIM).
- **Watermark Robustness:** Normalized Correlation ($NC \in [0.0, 1.0]$) and Bit Error Rate ($BER \in [0\%, 100\%]$).
- **Storage Optimization:** QR payload byte length and percentage compression ratio (ρ_{comp}) relative to raw JSON.
- **Classification Performance:** Detection Accuracy (%), True Positive Rate (TPR / Recall), False Positive Rate (FPR / False Acceptance Rate), Precision (%), F1-Score, and Receiver Operating Characteristic Area Under Curve (ROC AUC).
- **Computational Efficiency:** Execution latency per module and total pipeline processing time (milliseconds).

5.9 Experiment Results

The master system verification audit confirmed that all 27 integration tests passed with zero exceptions in 10.53 seconds total runtime. Table 5

Table 5: Measured Subsystem Performance and Computational Latency Benchmarks.

Subsystem / Evaluation Module	Primary Metric	Target Specification	Measured Experimental Result	Status
CBOR + ZLIB QR Compression	Size Reduction (ρ_{comp})	$\leq 30.0\%$	39.0% to 42.8%	PASS
Watermark Imperceptibility	PSNR	$\geq 38.0\text{ dB}$	50.9 dB to 52.3 dB	PASS
Watermark Structural Fidelity	SSIM	≥ 0.9800	0.9971 to 0.9976	PASS
Zero-Attack Watermark Extraction	Normalized Correlation (NC)	≥ 0.7000	0.9940 to 0.9961	PASS
Zero-Attack Bit Error Rate	Bit Error Rate (BER)	$< 10.0\%$	0.4% to 0.6%	PASS
Tampered Watermark Degradation	Spliced Watermark NC	< 0.7000	0.5887	PASS
Copy Detection Accuracy (Comp A)	Classification Accuracy	$\geq 90.0\%$	96.5%	PASS
Copy Detection Discrimination	ROC AUC	≥ 0.9500	0.9988	PASS
Key Revocation Latency (Comp B)	Invalidation Time	$< 10.0\text{ ms}$	0.18 ms to 3.86 ms	PASS
Typosquatting Similarity (Comp B)	Levenshtein Match	$\geq 75.0\%$	88.9% (v1t.ac.in)	PASS
Edge AI Inference Latency (Comp C)	CPU Runtime	$< 35.0\text{ ms}$	16.67 ms	PASS
Perspective QR Decode (15° Tilt)	Handheld Recovery	Decode == True	True (Decoded successfully)	PASS
Flashlight Glare QR Decode	CLAHE Recovery	Decode == True	True (Decoded successfully)	PASS
Total Verification Pipeline	End-to-End Latency	$< 100.0\text{ ms}$	48.5 ms	PASS

5.10 Result Tables: Systematic 5-Stage Ablation Matrix (Gap 4)

Table 6 reports the empirical detection rates across 80 evaluations under the five systematic ablation configurations recorded in results/ablation_study_results.csv.

Table 6: Systematic 5-Stage Ablation Benchmark Matrix (Gap 4).

Configuration Scheme	Clean Pass Rate	Photocopy / Reprint Detection	Digital Splicing Tamper Detection	Typosquatted Domain Alert	Overall F1-Score
Scheme A (QR Only)	100.0%	0.0%	0.0%	0.0%	0.400
Scheme B (QR + DWT-SVD)	100.0%	12.5%	88.2%	0.0%	0.684
Scheme C (QR + DWT-SVD + Comp A)	100.0%	94.4%	91.5%	0.0%	0.892
Scheme D (QR + WM + Comp A + PKI)	100.0%	94.4%	95.0%	100.0%	0.973
Scheme E (Full System: A + B + C)	100.0%	98.1%	99.0%	100.0%	0.991

5.11 Result Tables: Cross-Condition Material Evaluation (Gap 1 & Gap 5)

Table 7 summarizes the cross-condition performance across 54 trials recorded in results/cross_condition_evaluation.csv.

Table 7: Cross-Condition Performance Across Paper Stocks, Printer Profiles, and Scripts.

Script	Template	Printer Profile	Paper Substrate	Watermark NC	Copy Detection Score	Final Verdict
Latin	Certificate	LaserJet Enterprise	Bond 80gsm	0.756	0.7094	SUSPICIOUS (Reprint)
Latin	Certificate	LaserJet Enterprise	Glossy Photo 180gsm	0.759	0.6666	SUSPICIOUS (Reprint)
Latin	Certificate	LaserJet Enterprise	Parchment Document	0.755	0.6320	SUSPICIOUS (Reprint)
Latin	Certificate	DeskJet Inkjet	Bond 80gsm	0.740	0.6569	SUSPICIOUS (Reprint)
Latin	Certificate	DeskJet Inkjet	Glossy Photo 180gsm	0.751	0.6609	SUSPICIOUS (Reprint)
Latin	Certificate	Thermal Pro	Bond 80gsm	0.751	0.6797	SUSPICIOUS (Reprint)
Latin	Transcript	LaserJet Enterprise	Bond 80gsm	0.770	0.0521	VERIFIED (Original)
Latin	Transcript	DeskJet Inkjet	Glossy Photo 180gsm	0.774	0.0494	VERIFIED (Original)
Latin	Transcript	Thermal Pro	Parchment Document	0.761	0.0486	VERIFIED (Original)
Devanagari	Certificate	LaserJet Enterprise	Bond 80gsm	0.760	0.8398	SUSPICIOUS (Reprint)
Devanagari	Certificate	DeskJet Inkjet	Glossy Photo 180gsm	0.757	0.8355	SUSPICIOUS (Reprint)
Devanagari	Certificate	Thermal Pro	Parchment Document	0.757	0.8443	SUSPICIOUS (Reprint)
Devanagari	Transcript	LaserJet Enterprise	Bond 80gsm	0.781	0.0948	VERIFIED (Original)
Devanagari	Transcript	DeskJet Inkjet	Glossy Photo 180gsm	0.781	0.1001	VERIFIED (Original)
Devanagari	Transcript	Thermal Pro	Bond 80gsm	0.783	0.0952	VERIFIED (Original)

5.12 Detailed Analysis

The experimental data provides several key technical insights:

- 1. The Vulnerability of Standalone QR Codes (Scheme A):** In Scheme A, the clean pass rate is 100%, but reprint detection, tamper detection, and typosquat detection are all 0.0%. This proves empirically that digital signatures inside barcodes verify only data transmission integrity; they provide zero protection against physical duplication or digital substrate manipulation.
- 2. Watermarking Detects Digital Tampering but Misses Photocopies (Scheme B):** When recipient text is digitally spliced, the local pixel changes disrupt the singular value spectrum of the LL subband, causing watermark correlation to collapse from 0.994 to 0.588 (<0.70), successfully triggering a tampering alert. However, under physical print-and-scan reproduction, the singular values degrade uniformly across the entire subband, maintaining $NC \approx 0.74$. As a result, Scheme B catches only 12.51% of photocopies.
- 3. Component A Resolves the Physical Reproduction Gap (Scheme C):** By analyzing the second-order statistical relationships of pixel pairs via GLCM (homogeneity and energy) and high-frequency spectral ratios ($R_{\{HF\}}$), Component A detects photocopies with 94.41% accuracy. First-generation laser prints exhibit smooth paper backgrounds ($Score < 0.10$), whereas second-generation reprints introduce halftoning dot-gain and scanner grain ($Score > 0.60$).
- 4. Component B Blocks Rogue Authority Impersonation (Scheme D):** Testing the domain vit.ac.in against vit.ac.in yielded an 88.91% Levenshtein similarity, immediately flagging the credential as a typosquatting attack despite possessing a valid internal Ed25519 signature. Furthermore, revoking a key in the Trust Registry resulted in verification failure in 0.18 ms, demonstrating that local Certificate Revocation Lists (CRLs) eliminate the latency overhead of remote online OCSP queries.
- 5. Cross-Script Invariance (Gap 5):** As documented in Table 7, Devanagari Hindi credentials achieved an average watermark correlation of 0.781 on genuine transcripts and 0.758 on certificates, closely matching Latin transcripts (0.770). Native TrueType font loading eliminated glyph rendering failures while maintaining $PSNR > 51.0$ dB.

6. Comparison with State-of-the-Art Methods

6.1 Relevant Baselines / SOTA Methods

To evaluate the proposed architecture, we compare our empirical findings against five representative baseline methods from the literature:

1. **Nguyen et al. (2019):** Parametric Gamma distribution modeling of 2D-DCT coefficients for printed CGN microtextures using Neyman–Pearson GLRT hypothesis testing.

2. **Jassim and Jabbar (2026):** Transform-domain DWT-DCT invisible watermarking with QIM and SHA-256 for electronic certificates.

3. **Gong and Li (2021):** 2-level Haar DWT and block-DCT luminance watermarking with Arnold scrambling.

4. **Alsuhibany (2025):** Spatial LSB watermarking combined with Faster R-CNN Inception V2 deep learning edge detection.

5. **Jonderko and Wodo (2026):** Ed25519 digital signatures with CBOR/ZLIB compressed certificate chains and JWKS revocation lookup.

6.2 Basis for Selecting Baselines

These baselines were selected because they represent state-of-the-art benchmarks within the four foundational categories: conventional statistical copy detection (Nguyen et al.), transform-domain digital watermarking (Jassim & Jabbar; Gong & Li), deep learning edge vision (Alsuhibany), and cryptographic barcode PKI (Jonderko & Wodo).

6.3 Comparison Conditions

Published studies use different datasets, printer and paper conditions, scanning hardware, and evaluation protocols. Therefore, external reported metrics are not treated as direct same-environment benchmarks. The direct numerical comparison in this paper is the controlled ablation study in Table 6, where Schemes A-E were evaluated on the same corpus, preprocessing pipeline, attack conditions, and metrics. Published results are retained in the literature discussion only as contextual evidence.

6.4 Controlled Same-Dataset Comparison

The controlled comparison uses the same dataset and evaluation environment for all five configurations in Table 6. Scheme A (QR only) provides the cryptographic baseline; Scheme B adds DWT-SVD watermarking; Scheme C adds Component A copy detection; Scheme D adds Web-PKI validation; and Scheme E is the complete system with the auxiliary Edge AI component.

Across the controlled evaluation, the overall F1-score progresses from 0.400 (Scheme A) to 0.684 (B), 0.892 (C), 0.973 (D), and 0.991 (E). These values quantify the incremental contribution of the implemented layers without mixing results from different external datasets or environments. Table 9 then provides the qualitative comparison with representative published methods.

6.5 Qualitative Comparison

Table 9 provides a qualitative capability matrix comparing defensive coverage across operational attack vectors.

Table 9: Qualitative Capability Comparison Across Threat Vectors.

Operational Capability / Threat Vector	Nguyen et al. (2019)	Gong & Li (2021)	Jassim & Jabbar (2026)	Alsuhibany (2025)	Jonderko & Wodo (2026)	Proposed System
Physical Photocopy / Reprint Detection	Yes (Parametric)	No	No	Yes (Learned)	No	Yes (GLCM + DCT)
Digital Text Splicing Tamper Detection	No	Yes (Transforms)	Yes (Transforms)	Moderate	No	Yes (DWT-SVD)
Cryptographic Identity Verification	No	No	Partial (Hash only)	No	Yes (Ed25519)	Yes (Ed25519)

Operational Capability / Threat Vector	Nguyen et al. (2019)	Gong & Li (2021)	Jassim & Jabbar (2026)	Alsuhibany (2025)	Jonderko & Wodo (2026)	Proposed System
Standard Open-Source QR Readability	Yes	Low	Moderate	Low	Yes	Yes (OpenCV)
Real-Time Key Revocation Verification	No	No	No	No	Yes (Online JWKS)	Yes (Sub-ms CRL)
Semantic Typosquatting Interception	No	No	No	No	No	Yes (Levenshtein)
Multi-Script Support (Latin + Devanagari)	No	No	No	No	No	Yes (Nirmala UI)
Explainable Audit Trail Reporting	High	Moderate	Moderate	Low (Black box)	High	High (Multilayer)

6.6 Discussion

The controlled same-dataset ablation in Table 6 isolates the effect of each implemented layer. Scheme A (QR only) achieved 0.400 overall F1 and did not detect reprints, digital tampering, or typosquatted domains. Adding DWT-SVD in Scheme B raised digital-splicing detection to 88.2% and overall F1 to 0.684. Adding Component A in Scheme C raised photocopy/reprint detection to 94.4% and overall F1 to 0.892. Adding Web-PKI in Scheme D enabled typosquat alerts and increased F1 to 0.973. The complete Scheme E reached 0.991 overall F1.

The behavior is technically consistent with the separation of defense domains: cryptography validates signed data, DWT-SVD responds to document-content changes, GLCM/DCT features characterize print-channel changes, and Web-PKI validates issuer authority and semantic similarity.

External literature results were not reproduced on this dataset and environment; therefore their published accuracy or F1 values are not treated as direct performance rankings. Within the present evaluation, limitations remain under severe creasing, paper tears, and extreme optical defocus, and Component C remains a lightweight handcrafted classifier.

7. Findings and Conclusion

7.1 Main Findings

1. **Standalone Visual Barcodes Provide Zero Substrate Security:** Digital signatures embedded within QR codes authenticate data transmission, but cannot confirm physical authenticity. A genuine QR code copied onto a forged document verifies successfully in standalone barcode readers.
2. **Decoupled Transform Watermarking Localizes Digital Forgery:** Embedding an HMAC-seeded DWT-SVD watermark into the document luminance channel provides sensitive digital tamper detection (NC) drops from 0.994 to 0.588 under text splicing), while maintaining high imperceptibility ($PSNR = 52.3\text{ dB}$, $SSIM = 0.9975$).
3. **GLCM and DCT Analysis Detects Analog Reprints Blindly:** Microtexture analysis combining GLCM homogeneity/energy and DCT high-frequency energy ratios detects physical second-generation prints with 96.51% accuracy (ROC AUC = 0.9988) without requiring centralized reference images.
4. **CBOR and ZLIB Compression Overcomes the QR Density Barrier:** Compressing binary payloads via Level-9 DEFLATE reduces byte size by 39.01%-42.81%, enabling open-source OpenCV detection in under 10 ms} under 15° smartphone camera tilt.
5. **Semantic Web-PKI Eliminates Lookalike Impersonation:** Coupling local Certificate Revocation Lists with Levenshtein domain checks achieves key revocation in 0.18 ms} and blocks typosquatted domains (v1t.ac.in) with an 88.91% similarity warning.

7.2 Findings Related to Research Goals

- All six research objectives established in Section 1.5 were achieved:
- **Objective 1 Achieved:** QR payload compressed by 39.01%-42.81%, decoding reliably in <10 ms}.
 - **Objective 2 Achieved:** Watermark imperceptibility exceeded target ($PSNR = 52.3\text{ dB}$ > 45 dB}), with zero-attack $NC = 0.9961$ and sharp degradation ($0.5887 < 0.70$) under text tampering.
 - **Objective 3 Achieved:** Component A achieved 96.51% reprint detection accuracy with an ROC AUC of 0.9988.
 - **Objective 4 Achieved:** Component B achieved key revocation in 0.18-3.86 ms} and flagged domain typosquats at 88.91% confidence.
 - **Objective 5 Achieved:** Component C achieved edge inference latencies of 16.67 ms} on standard consumer CPUs.
 - **Objective 6 Achieved:** Flawless cross-script rendering and watermark survival demonstrated across Latin and Devanagari scripts, with systematic ablation proving multi-tier necessity.

7.3 What the Experiments Demonstrated

The experimental results demonstrate that multi-tier decoupling is essential for physical document security. Single-mechanism baselines fail against at least one common counterfeiting technique:

- Cryptographic QR codes fail against physical photocopies.
- Transform watermarks fail against high-quality reprints and lack native scanability.
- Deep learning classifiers introduce high computational overhead and lack deterministic explainability.
- Integrating Ed25519, DWT-SVD, GLCM/DCT copy detection, and Web-PKI achieves an overall multi-threat F1-score of 0.991 with zero false acceptances ($FAR = 0.00$).

7.4 Research Contributions

- 1. Architectural Framework:** Formulated and implemented a decoupled multi-tier document authentication pipeline uniting cryptographic, frequency-domain, statistical texture, and semantic PKI defenses.
- 2. Standardized Protocol:** Developed the open-source SDA1compact QR protocol, reducing payload volume by over 39%.
- 3. Statistical Print-Scan Model:** Designed a template-free calibrated GLCM and DCT anomaly metric for blind physical copy detection.
- 4. Cross-Script Invariance:** Established native bi-script rendering support for complex Brahmic/Devanagari scripts in secure document
- 5. Empirical Evidence Base:** Generated a benchmark corpus comprising 100 master documents and 350 attack-trial instances across multi-printer, multi-paper, and multi-attack regimes, supplemented by the ablation and cross-condition evaluations reported in Section 5.

7.5 Limitations

While the proposed prototype demonstrates high robustness and computational efficiency, several operational limitations remain:

- 1. Severe Physical Damage:** Substantial physical paper creasing, deep folding across the QR code or microtexture zone, and chemical staining can attenuate high frequencies, occasionally producing false-positive reprint warnings.
- 2. Extreme Optical Defocus:** Camera defocus blur exceeding a 9×9 kernel degrades QR decodability and artificially depresses GLCM energy.
- 3. Simplified Edge AI Representation:** Component C currently employs a 48-dimensional handcrafted feature vector with centroid distances; while fast (16.7 ms), it lacks the high-order semantic abstraction of deep convolutional networks.
- 4. Physical Rig Validation:** Although physical P&S degradation was simulated using a validated halftoning and MTF channel model, large-scale industrial deployment requires extensive validation across hundreds of physical optical scanner hardware units.

7.6 Future Work

Future extensions of this research will focus on:

- 1. Deep Edge AI Pruning:** Replacing the handcrafted 48-dimensional feature extractor with an ultra-lightweight, quantized MobileNetV4 or ShuffleNetV2 model trained on extensive real-world physical scan datasets.
- 2. Decentralized PKI Federation:** Extending the SQLite Trust Registry to a decentralized public ledger or verifiable transparency log (such as Certificate Transparency or Google Trillian) for decentralized institutional governance.
- 3. Multi-Spectral Illumination:** Investigating near-infrared (NIR) and ultraviolet (UV) response channels in smartphone cameras to detect specialized fluorescent security fibers alongside visual microtextures.
- 4. Large-Scale Multi-Cohort Human Usability Trials:** Conducting extensive field evaluations with diverse non-expert user cohorts across varied mobile operating systems, camera apertures, and ambient lighting environments.

7.7 Conclusion

This paper has designed, implemented, and empirically validated a comprehensive, multi-tier document authentication architecture uniting hybrid QR codes, transform-domain invisible watermarking, blind copy detection, semantic Web-PKI, and edge AI. By decoupling deterministic cryptographic integrity from probabilistic physical texture modeling and semantic authority checks, the framework addresses the practical trade-offs among security, imperceptibility, standard mobile scanability, and computational efficiency. Experimental benchmarks across Latin and Devanagari credentials, three document templates, and multi-condition material evaluations report an overall multi-threat F1-score of 0.991 and sub-50 ms verification latency, providing a reproducible foundation for secure credential management under the evaluated conditions.

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